

Quantum Interfaces problem set

Problem 1) Atom-light interaction along waveguides (13 points)

A two-level atom couples to both free-space and a guided (evanescent) optical mode of a nanofiber

- (a) (1 point) If the free space decay rate (without a waveguide) is $\Gamma_0 = 2\pi \times 6$ MHz and with $\beta = \Gamma_{1D}/\Gamma_{\text{Tot}} = 0.3$ what is the lifetime of the atom?
- (b) (1 point) Why is a subwavelength mode area crucial to reaching high β ?
- (c) (1 point) Do any experimental systems exist where β is close to 1?
- (d) (6 points) Consider two systems A and B with atoms coupled to the evanescent mode of a nanofiber waveguide.

System A) Uses rubidium atoms located 150nm from the surface of an optical nanofiber (refractive index 1.46) and operates on the 780nm $5S_{1/2} - 5P_{3/2}$ Rb transition $\Gamma_0 = 2\pi \times 6$ MHz. The nanofiber is 400nm in diameter.

System B) Uses ytterbium atoms located 150nm from the surface of an optical nanofiber (refractive index $n=1.46$) and operates on the 1388nm Yb $^3P_0 - ^3D_1$ transition $\Gamma_0 = 2\pi \times 309$ kHz. The nanofiber is 600nm in diameter.

Assume the $1/e^2$ waist diameter ($2W_0 = \text{MFD}$, $I(r) = I_0 \exp\left(-2\frac{r^2}{w^2}\right)$) is given by:

$$\text{MFD} \approx a \left(0.65 + \frac{1.619}{V^{3/2}} + \frac{2.879}{V^6} \right) \quad \text{where} \quad V = \frac{\pi a}{\lambda} \sqrt{n^2 - 1}$$

For each system, estimate β . Justify your reasoning/calculation.

- (e) (4 point) Collective effects in defect free subwavelength arrays along waveguides have been predicted to strongly enhance or suppress emission and atom light interaction. What practical challenges exist to creating these conditions? Would System A or System B (from part d of this question) offer any practical advantages/disadvantages? Justify your answer.

Problem 2) Atom-light interaction in optical cavities (15 points)

Consider a Fabry-Perot cavity with length $L=1$ mm and two lossless mirrors both with reflectivity 99.99% (intensity reflection).

- (a) (2 points) Cavity properties:
 - (i) Calculate the free spectral range, the finesse of the cavity.
 - (ii) (+2 bonus points) Calculate the range of mirror radius of curvatures needed to produce a stable cavity mode. Choose a sensible mirror radius of curvature for good interfacing to neutral atoms in tweezer traps or ions. Justify your answer.
 - (iii) If the cavity is resonant what will be the steady state transmitted power after both the 99.99% reflective mirrors? (incident power = P_0)
 - (iv) What will be the circulating power in the cavity in terms of the incident power P_0
 - (v) Calculate the cavity decay rate κ .

Now place a rubidium atom inside the cavity mode. Assume $g = 2\pi \times 20$ MHz and $\Gamma_0 = 2\pi \times 6$ MHz.

- (b) (1 point) What is the single atom cooperativity of the system? (Use $C_1 = g^2/(2\kappa\gamma)$, $\gamma = \Gamma_0/2$ is the atomic dipole decay rate)

Is this strong coupling, Purcell or weak coupling regime? Explain.

- (c) (3 points) What possible sources of mode mismatch could occur coupling light into/out of this cavity? Can you estimate the magnitude/importance of each source. Justify your answer
- (d) (1 point) If this cavity is used as a phase gate producing a reflection phase shift near π how will the mode mismatch efficiency ξ affect the achievable phase contrast?
- (e) (1 point) Is there a different cavity design that could reduce the mode mismatch error?
- (f) (1 point) The cavity is shortened, decreasing mode volume by factor 4 (tighter mode) but increasing κ by factor 2. Predict the new cooperativity and discuss whether the modification helps or hurts for high-fidelity atom-photon gates.
- (g) (2 points) Imagine that the cavity mirror curvatures are chosen to give a $1/e^2$ cavity waist radius of 1 μm and this cavity waist is overlapped perfectly with the atom in tweezer. Would this affect the decay rate of the atom? Justify your answer.
- (h) (4 points) A cavity can be formed using traditional mirrors, by grinding, polishing and coating optical fibers, through nanostructures or through whispering gallery modes. Describe advantages and disadvantages of each approach in terms of achievable cooperativity, modematching, coupling to atoms/ions/NV centers for each technology.

Problem 3) Interfacing Quantum Processors (2 x 13 points)

Congratulations Professor! You have just been appointed to lead the newly created Quantum Processor Networking Division of the Research Center for Critical Issues within Academia Sinica.

Develop a proposal to interface two (different) of Taiwan's matter based quantum processors (you may choose trapped ions, trapped neutral atoms or superconducting qubits). You may use the already constructed Taiwan quantum communication network which can provide high bandwidth (10MHz) heralded telecom band photons (1.3-1.6 μm) with high fidelity at both processor nodes. All you have to do is construct the interface from these fibers to the two processor types:-)

For each type of quantum processor that you will interface to:

- (3 points) Describe the processor (eg atom/ion type) that you will interface to, the main technical challenges and realistic current state of the art performance targets that you will aim for (eg bandwidth/entanglement rates, added noise/fidelity). Cite papers to support your claims.
- (1 point) If you achieve these (state of the art) performance targets would they significantly degrade the networked quantum computer performance compared with state of the art single processor computation? What about for an interface to different processor types?
- (4 points) Sketch detailed diagram(s) describing the physics of how the telecom photon will be interfaced with the matter based qubit. You are encouraged to copy diagrams from papers as long as you cite the relevant papers. Describe the process step by step to clearly explain how the quantum information is transferred from the photon to the atom or vice versa. Be detailed, if it is an atom describe the energy levels/photon polarizations.
- (3 points) Describe alternative design choices/architectures and say why you chose this approach. Cite the relevant papers.
- (2 points) Describe key technical challenges that limit system performance. Describe how this could be improved. Cite papers!

If anything is not clear, ask! Contact: Shayne Bennetts, s.p.bennetts@g.iams.sinica.edu.tw

All materials (Slides/Homework problems) can be downloaded from our website (including any updates/corrections) www.iamsquantum.github.io/TIGP.html